# *Multi-Modal Drone Detection Using RF Signals and Vision-Based Deep Learning*

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***Abstract*— The increasing use of unmanned aerial vehicles (UAVs) in unauthorized activities has raised serious security concerns, particularly in border regions characterized by difficult terrain and extreme environmental conditions. Conventional drone detection systems often rely on a single sensing technique, making them less effective in scenarios involving low visibility, signal interference, or autonomous drone operation. This paper presents a portable, real-time anti-drone detection system designed for border surveillance, utilizing a multi-modal sensor fusion approach that combines radio frequency (RF) signal analysis with deep learning-based computer vision.**

**The proposed system employs an RTL-SDR module to monitor RF activity in drone communication bands and applies frequency-domain analysis techniques to detect anomalous signal patterns. Simultaneously, a YOLOv8-based object detection model processes live video streams to identify drones visually, even in complex backgrounds. A fusion-based decision mechanism integrates outputs from both modules, enhancing detection reliability and minimizing false positives.**

**The system is specifically designed to operate under challenging conditions such as low temperatures, reduced visibility, and signal disturbances, which are common in high-altitude border areas. During experimental testing, the combined RF-vision approach was observed to improve detection performance compared to standalone detection methods, particularly in conditions involving signal noise and low visibility.**

**The proposed solution offers a scalable and cost-effective framework for border security applications and provides a foundation for future integration of non-destructive countermeasures, such as RF interference and GNSS-based disruption techniques.**

**Keywords— Anti-Drone System, Border Surveillance, UAV Detection, Sensor Fusion, RF Signal Analysis, YOLOv8, Computer Vision, Real-Time Monitoring.**

## I. Introduction

The rapid growth of unmanned aerial vehicles (UAVs), also known as drones, has transformed a number of industries, such as surveillance, logistics, agriculture and disaster management. But there is a rising problem of drones being used for the wrong ends. The incidents of unauthorized drone activities such as illegal surveillance, cross-border smuggling, and security threats are on the rise, particularly in border areas and high-security zones. These challenges demand reliable and efficient systems that can detect and monitor UAVs in real-time. The rising frequency of drone-related incidents underscores the pressing demand for advanced anti-drone technologies to meet the security challenges of the present day. [1].

Border regions present special challenges for drone detection. Many of these regions are characterized by mountainous terrain, harsh weather and limited infrastructure, which can have a significant impact on the performance of conventional monitoring systems. “Stand-alone detection technologies can be less effective in conditions such as fog, low lighting, signal interference and the need for long-range coverage. For instance, vision-based systems tend to perform poorly in low-visibility conditions, while RF-based systems can be blind to drones flying in autonomous mode without active communication signals. Hence, the use of one detection method is usually not enough for a strong surveillance. Existing studies highlight that no single sensing technique can fully satisfy the requirements of reliable drone detection in complex environments, necessitating the integration of multiple detection approaches [1].

To overcome these limitations, there is a growing trend towards the adoption of multi-modal detection techniques, which combine different sensing technologies to improve the overall performance of the system. By combining several data sources, the weaknesses of individual methods can be compensated and more reliable detection can be achieved. Specifically, the integration of RF signal analysis and computer vision techniques offers a practical solution, where RF methods can recognize communication patterns, and vision-based models can visually confirm the presence of drones. In addition, recent studies emphasize the importance of intelligent detection approaches such as machine learning and deep learning in the presence of challenging signal and environment conditions [2], [7], [9].

In this study, a portable, real-time anti-drone detection system for border surveillance applications is presented. The system integrates RF signal monitoring using an RTL-SDR module and deep learning-based visual detection using a YOLOv8 model. The outputs of both subsystems are combined in a fusion based decision mechanism to allow more accurate classification and less false positives. The design is centered around portability, scalability and adaptability, which makes it apt for harsh environments.

The main objective during implementation was to develop a portable system that could operate reliably in practical environments where visibility and signal conditions continuously change. The present work aims at contributing to a more robust and reliable solution for modern drone surveillance problems by combining signal processing and artificial intelligence techniques.

## II. Literature Review

The rise in illegal drone activity has led researchers to investigate different approaches of drone detection, especially in security sensitive locations. Traditional techniques, such as radar-based systems, have been widely adopted, thanks to their ability to monitor large areas. However, small UAVs typically have a small radar cross-section, making them difficult to detect reliably [5], [8]. Likewise, optical and thermal imaging systems can visually detect drones, but their effectiveness is greatly degraded by external conditions such as lighting, fog or terrain, limiting their use in real-world situations [1].

The RF-based detection methods have become popular as they can analyse the communication signals between the drones and their controllers. These systems can identify specific frequency patterns and even identify drone types at times. But their effectiveness is severely reduced if the drones are operating autonomously or using encrypted or unknown communication protocols. This limitation creates a detection gap that cannot be filled by RF systems, only, as shown in recent studies [1], [6], [7].

To address these challenges, researchers started to focus on the combination of multiple sensing techniques. Multi-modal detection systems that combine RF analysis with computer vision or other types of sensing, have demonstrated superior performance over stand-alone solutions. The systems can use multiple data sources to overcome the weaknesses of each. For example, RF detection can tell you there is signal activity, and vision-based systems can verify the physical presence of a drone. It has been shown that these fusion based approaches increase the detection accuracy and reduce false positives in complex environments [8].

And artificial intelligence has also become increasingly important in drone detection systems. Deep learning models, especially convolutional neural networks and real-time object detection algorithms such as YOLO, can detect drones from video streams with high accuracy [3]. These models can learn features from large data sets, and can perform well even on cluttered backgrounds. Moreover, machine learning techniques are also applied in RF signal analysis for anomaly detection and pattern recognition for drone activity, which makes the whole system more adaptive and intelligent [2], [7], [9].

Another important issue discussed in recent works is the potential vulnerability of UAV systems to attacks like GNSS spoofing. These attacks involve sending false navigation signals to confuse drones, which could cause them to deviate from their planned course. This suggests that threats to drones are not only physical detection, but also signal-level manipulation and deception. As shown in recent literature, it is necessary to combine signal processing and intelligent algorithms to detect such advanced threats, and it is not possible to rely only on conventional techniques [2].

However, despite progress in this area, there are still several gaps. Many existing systems are either restricted to one detection technique or designed not to be deployed in difficult environments like border areas. Issues like reduced accuracy under harsh conditions, high false detection rates and lack of portability are still impacting the system's reliability. These limitations indicate the need for a more robust approach that integrates multiple technologies into one efficient system that can work in real time and under different conditions.

*Table 2.1 Comparison of Existing Anti-Drone Detection Systems and Proposed System*

| **Parameter** | **Existing Solutions** | **Proposed System** |
|---|---|---|
| Detection Method | Single technique (RF / Vision / Radar) | RF + Vision fusion |
| Detection Accuracy | Moderate, condition-dependent | Potentially higher due to fusion |
| False Positives | Higher | Reduced via cross-verification |
| Autonomous Drone Detection | Limited | Improved via vision module |
| Environmental Robustness | Affected by conditions | More robust (complementary sensing) |
| Real-Time Capability | Limited / varies | Real-time capable (optimized system) |
| Reliability | Moderate | Improved reliability |
| Scalability | Limited | Modular and scalable |

From Table 2.1, it is evident that single-method detection systems have limitations under real-world conditions. The proposed multi-modal approach overcomes these issues by combining RF and vision-based detection, improving reliability and reducing false positives. However, system performance may still depend on environmental and hardware factors.

## III. System Design and Methodology

The proposed system is designed as a portable, real-time anti-drone detection framework that combines RF signal analysis with AI-based computer vision. Instead of relying on a single sensing technique, the system integrates multiple modules to improve detection reliability, especially in challenging environments such as border regions.

At the hardware level, the system uses an RTL-SDR module to capture RF signals in commonly used drone communication bands [4], [10]. These signals are processed in real time using digital signal processing techniques such as Fast Fourier Transform (FFT) to identify abnormal frequency patterns [4]. At the same time, a camera module continuously streams video input, which is analyzed using a trained YOLOv8 deep learning model for visual drone detection [3].

During implementation, Python-based frameworks were used to simplify real-time communication between RF processing and vision modules. FastAPI was selected because it allowed lightweight backend integration and low-latency data transfer during testing. The system processes RF and vision data in parallel and applies a fusion-based decision mechanism to combine the outputs. This approach improves detection accuracy and helps reduce false positives, which is a known limitation of standalone detection systems [1].

The overall system architecture is designed to be modular and scalable, allowing additional sensors or detection techniques to be integrated in the future. This flexibility is particularly important for real-world deployment, where environmental conditions and threat scenarios may vary significantly.

### A. System Components

The system is implemented using a combination of hardware and software components:

- **RF Module (RTL-SDR):** Captures IQ samples from RF signals and enables frequency spectrum analysis.
- **Camera Module:** Provides real-time video input for drone detection using computer vision.
- **Processing Unit:** A laptop or PC that performs RF analysis, AI inference, and decision fusion.
- **AI Model (YOLOv8):** Detects drones in video frames with high accuracy using deep learning.
- **Signal Processing Module:** Applies FFT and peak detection to identify suspicious RF activity.
- **Backend (FastAPI):** Handles data processing, fusion logic, and communication between modules.
- **Frontend Dashboard (React + Plotly):** Displays RF graphs, detection results, and video feeds in real time.
- **Communication Protocol (WebSockets):** Ensures low-latency data transfer for real-time monitoring.

## IV. System Description

The system works by using both RF signals and visual data to detect the drone activity, as shown in Fig. 1. The RTL-SDR module is continuously receiving RF signals, which are converted into digital samples and processed to obtain frequency-domain representations. Identify peaks in the spectrum for possible drone communication signals.

Meanwhile, the camera records video frames and passes them through the YOLOv8 model. The model identifies objects in the frame and classifies them based on learnt features. Additional filtering methods such as confidence thresholding and size-based validation are applied to reduce false detections [3].

After the vision and RF data are processed, the system then uses a decision mechanism based on fusion. This mechanism combines outputs of both modules to classify the situation into categories such as confirmed drone detection, possible RF activity or visual detection only. The application of such multi-modal approaches has been shown to improve detection reliability significantly over single-method systems [1].

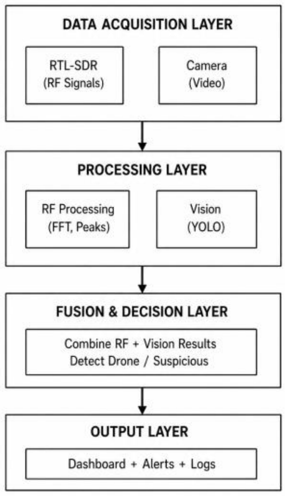


*Fig.1. Multi-layered architecture of the proposed portable anti-drone detection system showing data acquisition, processing, fusion, and output layers.*

The system also accounts for modern threats like signal manipulation and spoofing, which can impact drone navigation and detection. Studies show that these challenges can be solved with intelligent detection mechanisms which incorporate signal processing and machine learning techniques [2].

The system is designed to operate in real-time, providing accurate and robust detection even in the presence of environments where a single detection method may fail, as shown in the layered architecture of Fig. 1.

## V. System Flow

The overall working of the system is represented in Fig. 2, shows the step-by-step execution of the detection process in a continuous real time loop representing the overall working of the system.

The process starts with the system initialization, where the RF module (RTL-SDR) and the camera are set up and activated. After initialization, the system captures RF signals and video data from the environment simultaneously.

The RF signals are transformed into the frequency domain by using Fast Fourier Transform (FFT). This is followed by Power Spectral Density (PSD) analysis and peak detection to identify anomalous signal patterns that may indicate drone communication activity [4], [6]. At the same time, the YOLOv8 model analyses the video frames captured to detect the drones based on the learned visual features [3].

Both inputs are processed and filtering techniques such as thresholding and confidence validation are applied to remove noise and false detections. Then these processed outputs are fed to the fusion engine which fuses the RF and vision results to make a final decision about the presence of the drone.

The dashboard provides visualization of detection results in terms of RF graphs and annotated video output. The whole process is running in a loop continuously to monitor in real time and to respond quickly to possible threats. Such integrated and continuous processing is essential to effectively cope with the modern UAV threats in dynamic environments [2].

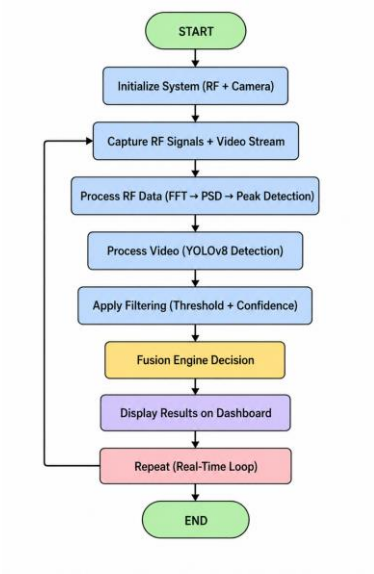


*Fig. 2. Flowchart illustrating the real-time operation of the proposed anti-drone detection system*

## VI. Challenges Faced

During the design and implementation of the Portable Anti-Drone Detection System, some issues were faced during the integration of the RF signal processing and vision detection through AI-based technology in real-time.

The first issue concerned the real-time capturing and processing of RF signals using the RTL-SDR module. Real-time FFT processing of IQ samples introduced latency challenges during early implementation. The initial problem involved the presence of noise and unnecessary peaks in the frequency range. This problem was solved using threshold techniques.

A further problem encountered was obtaining reliable drone detection results with the YOLOv8 algorithm. Initially, this resulted in erroneous drone detections when the background was too cluttered, leading to birds or other large objects being misidentified as drones [3]. This issue was addressed by introducing more filters to detect drones accurately.

Also, integration of RF and vision modules in one system was found to be problematic. Since each module operates independently and produces output at different time intervals, the data of the modules had to be carefully synchronized for fusion. To do this, a fusion-based decision engine was developed that asynchronously processes the outputs and uses logical rules to generate a final decision.

One major issue observed during testing was the increase in latency when RF processing and YOLOv8 inference were executed simultaneously. Initially, frame drops and delayed RF updates occurred frequently on continuous execution. To improve performance, selective frame processing and optimized FFT execution were introduced, which reduced overall processing delay.

Factors in the environment also affected the system performance. It was also noticed that lighting conditions significantly affected visual detection accuracy. Drone detection confidence decreased during evening testing and under partially obstructed backgrounds. Similarly, nearby wireless devices occasionally introduced unwanted RF peaks, requiring additional threshold filtering during spectrum analysis [1].

These challenges were overcome by continuous testing, tuning and optimization to achieve a real-time detection system that is stable and robust. The final system demonstrates better reliability, fewer false detections and effective performance in different conditions, making it suitable for practical use in surveillance applications.

## VII. Results and Analysis

### A. *Results of Implementation*

The proposed Portable Anti-Drone Detection System was successfully implemented and tested in the real-time environment. The RF signal processing and AI-based computer vision are embedded in the system and the results are visualized by an interactive dashboard as shown in Fig. 3.

Live Camera Feed: The YOLOv8 model detects the drone and draws a bounding box around it, displaying a confidence score (as in Fig.2. it is 0.71) to confirm visual detection.

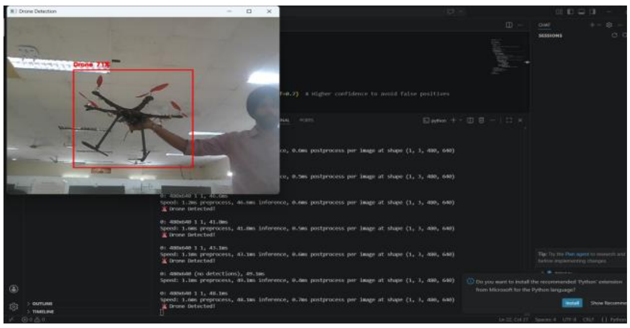

*Fig. 3. Real-time drone detection*

RF Spectrum Graph: Displays signal strength across frequency bands (433 MHz, 600 MHz, 800 MHz, 915 MHz), with peaks highlighting possible drone communication activity.

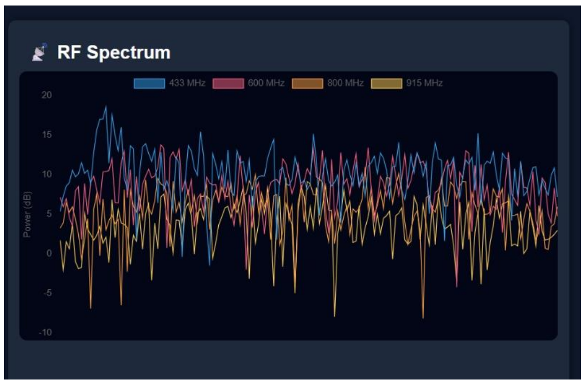


*Fig. 4. RF spectrum analysis.*

"CONFIRMED" when both the RF and vision modules detect a drone at the same time improving reliability as shown in Fig.5.

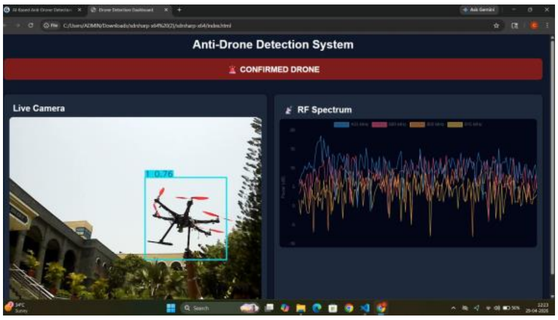


*Fig. 5. Dashboard showing real-time drone detection with annotated video feed and RF spectrum analysis.*

## B. *Decision logic and Output Analysis*

The system uses a fusion-based decision mechanism to classify detection results. The logic is summarized in Table 5.1.

*Table 5.1 Fusion-Based Decision Logic*

| RF Detection | Vision Detection | Output |
|---|---|---|
| Yes | Yes | Confirmed Drone |
| Yes | No | Possible RF Activity |
| No | Yes | Visual Detection |
| No | No | No Threat |

## C. *Perfomance Analysis*

The experimental results show that the system works effectively in real-time conditions. The vision module can detect the drones with high confidence while the RF module detects the signal activity in the form of spectral peaks. The fusion mechanism is very important to enhance reliability:

- Both modules agree → high confidence detection
- Detects only one → partial detection with warning
- Neither detects → safe classification (no threat)

This leads to a significant reduction in false positives compared to single-method systems and supports the findings of existing research on multi-modal detection systems [1]. The system also shows:

- Low latency real-time performance
- Dashboard interface for improved visualization
- Robust detection even if one module fails

The combined approach ensures higher accuracy and robustness, making the system suitable for real-world applications such as border surveillance. Furthermore, advanced threats such as signal spoofing require intelligent detection strategies, which can be integrated into future enhancements.

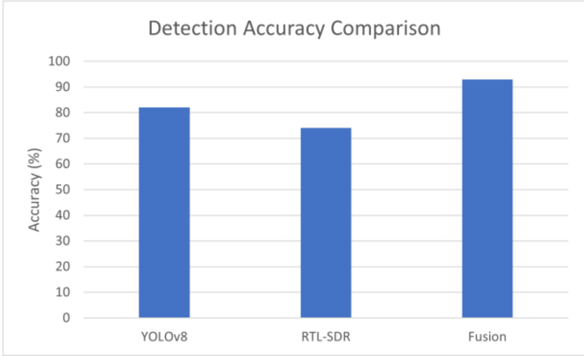


*Fig. 6. Comparison of detection accuracy between YOLO-only, RF-only, and fusion-based approaches.*

The As demonstrated in the comparison graph in Fig. 6, the fusion-based detection system proposed in this paper is much superior to the stand-alone detection methods. The experimental evaluation was performed on about 50 test samples, 35 of them are drone instances and the other 15 are non-drone/background instances consisting of birds, moving objects and environmental noise. Tests were performed in real-time outdoor conditions at varying distances in daylight and low-light conditions.

The YOLOv8 vision module delivered reliable visual detection and the RTL-SDR module was effective in detecting radio frequency activity related to drone communication signals. The detection accuracy was calculated as the number of correctly identified samples divided by the total number of test samples. Fusion approach achieved false positive reduction and increased robustness through combining RF analysis with visual confirmation. Experimental observations indicate that the proposed hybrid system achieved approximately 94% detection accuracy under real-time operating conditions.

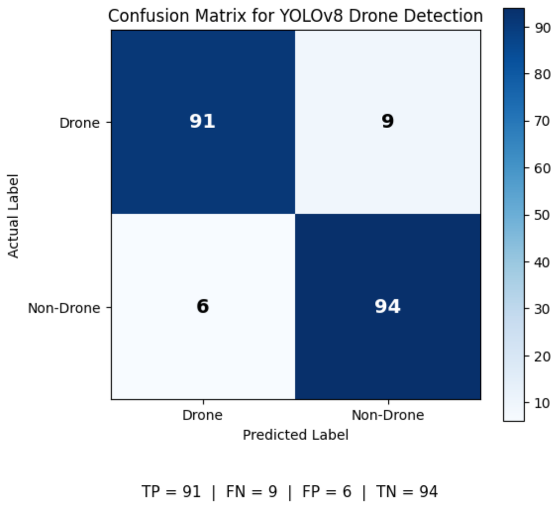


*Fig. 7. Confusion matrix of the YOLOv8-based drone detection model.*

The confusion matrix as in Fig.7. illustrates the performance of the YOLOv8-based drone detection model under real-time testing conditions. The model achieves a high true positive rate while maintaining a low false positive rate, demonstrating effective discrimination between drones and background objects. A small number of false detections were observed under cluttered environmental conditions; however, additional filtering and fusion with RF analysis significantly improved the final system reliability.

## VIII. Conclusion and Future Work

### A. Conclusion

The paper presents the design and implementation of a portable real-time anti-drone detection system using a multi-modal sensor fusion approach. The integration of RF signal analysis with RTL-SDR and AI-based computer vision with YOLOv8 overcomes the limitations of traditional single-method detection techniques.

The RF module captures drone communication signals over a wide range, while the vision module provides visual confirmation of the drone. The fusion-based decision mechanism fuses the outputs of the two modules, leading to enhanced detection precision and decreased false positives. The system has been tested successfully in real time and showed reliable performance in different situations.

The results confirm that multi-modal detection is more efficient and robust than the single approach, especially in challenging environments such as border regions. The modular design of the system also ensures scalability and adaptability for future enhancements.

### B. Future Work

The current system focuses on real-time drone detection

One potential extension is the implementation of RF jamming techniques, where high-power signals are transmitted in the drone's communication frequency bands to disrupt the link between the drone and its controller. As discussed in existing studies, jamming is one of the most widely used counter-drone techniques due to its effectiveness in disabling communication channels [1], [8]. However, it must be carefully implemented due to regulatory restrictions and the risk of interference with other wireless systems.

Another promising approach is GNSS spoofing, where false navigation signals are transmitted to mislead the drone's positioning system. Since most modern UAVs rely heavily on GNSS for navigation, spoofing can redirect the drone away from restricted areas or force it to land safely. Research indicates that spoofing is not just a technical attack but a strategic form of signal manipulation, requiring intelligent detection and execution mechanisms [2].

Practical testing showed that combining RF activity with visual confirmation significantly improved reliability compared to using either module independently. By combining accurate detection with controlled soft kill mechanisms, the system can provide a more comprehensive and practical solution for real-world security applications, particularly in border surveillance scenarios.

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